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Team Solo 2

CYB 310: Malware Analysis

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Executive Summary

Objective

This report analyzed a malicious Microsoft Excel macro-enabled file named tn4598151.xlsm, identified by SHA-256 hash 500856ee3fc13326cad564894a0423e0583154ef10531de4ab6e6d5df90d4e31. The sample is best classified as a malicious Office macro document functioning as a first stage downloader/dropper. Its purpose appears to be initiating code execution through Excel macros, creating a local staging directory, and attempting to prepare a secondary DLL payload named test.dll.

Key Findings

Static analysis provided the strongest evidence of malicious intent. The workbook contained an embedded VBA project file named vbaProject.bin, confirming macro code was present. Macro analysis identified a Workbook_Open routine, meaning the macro was designed to run automatically when the workbook opened. The macro also checked system environment information, created a think-cell directory under the user's local application data path, and referenced %LocalAppData%\think-cell\test.dll. Manual workbook inspection showed a fake Office 365 themed message instructing the user to click "Enable Editing" and "Enable Content." This social engineering lure was intended to convince the user to allow macro execution. The workbook also used CONCATENATE formulas and scattered command fragments to hide the final command structure from casual inspection. Dynamic analysis confirmed that the macro executed part of its staging behavior. In the Windows 10 VM, Excel created the think-cell directory, matching the macro code. However, the full second stage payload chain was not

observed. The analysis did not confirm successful creation or execution of test.dll, and Wireshark did not capture a clear outbound malware callback. ANY.RUN supported that Excel execution, but the sandbox appeared to enter a Windows error path before completing the payload chain.

Risk and Impact

The main organizational risk is user driven macro execution. If a user opens the workbook and enables content, the macro can begin staging malicious activity on the endpoint. In a real environment, this type of file could be used for initial access, after which additional payloads may be downloaded or staged. Because the final payload was not recovered, the exact end impact cannot be fully confirmed, but the document's behavior is clearly malicious.

Defensive Recommendations

Recommended defenses include blocking or quarantining the known SHA-256 hash, restricting macro-enabled Office attachments from unknown senders, and disabling macros for untrusted documents. Endpoint detection should monitor for Excel creating folders in user-writable locations, writing DLL files, or spawning processes such as rundll32.exe, cmd.exe, or powershell.exe. Security teams should also hunt for %LocalAppData%\think-cell and %LocalAppData%\think-cell\test.dll.

Limitations and Confidence

The primary limitation of this analysis was incomplete second stage payload execution. Static analysis confirmed malicious macro logic and dynamic analysis confirmed local staging behavior, but the payload test.dll was not recovered or executed during the observed runs. Confidence is high that the document is malicious and performs first stage macro based staging behavior. Confidence is lower regarding the final payload's full capabilities because the second stage DLL was not recovered and analyzed. Overall, the sample should be treated as a malicious Excel macro downloader/dropper with confirmed initial execution behavior and unconfirmed final payload execution.